Research Projects, Publications And Internship:

Bachelors Projects Title: Optimized LIF Neuron with Memristor-Based Synapses and STDP Learning for Energy-Efficient Neuromorphic Systems.

Ovearview:

- Designed and implemented an optimized Leaky Integrate-and-Fire (LIF) neuron model integrated with memristor-based synapses to emulate brain-inspired computing systems.
- Incorporated Spike-Timing Dependent Plasticity (STDP) learning mechanism to enable adaptive and biologically realistic synaptic weight updates.
- Focused on optimizing circuit design for low power consumption and energy efficiency, making it suitable for next-generation neuromorphic hardware applications.
- Gained hands-on experience in neuromorphic system design, circuit-level optimization, and understanding the intersection of VLSI design with artificial intelligence hardware.
- Developed strong skills in modeling, simulation, and analyzing spiking neural networks, along with insights into emerging technologies like memristors for scalable and efficient computing.

Publications:

- Our project work has been submitted for publication in a Scopus-indexed journal, demonstrating its research relevance and technical contribution to the field of neuromorphic engineering.
- The study presents an energy-efficient neuromorphic system using an optimized Leaky Integrate-and-Fire (LIF) neuron with memristor-based synapses and Spike-Timing Dependent Plasticity (STDP) learning mechanism.
- This work highlights our ability to contribute to ongoing research in advanced VLSI systems and brain-inspired computing architectures.
- The publication process has strengthened my understanding of research methodology, technical writing, and academic presentation.
- It reflects my commitment to pursuing research-oriented higher studies in VLSI design and emerging computing technologies.

Description:

Designed and analyzed an optimized Leaky Integrate-and-Fire (LIF) neuron model integrated with memristor-based synapses. Implemented Spike-Timing Dependent Plasticity (STDP) learning to achieve energy-efficient neuromorphic computation.

Key Features:

- Energy-efficient neuromorphic architecture
- Memristor-based synaptic design
- STDP learning mechanism implementation
- Optimized neuron model for low-power operation

Tools Used:

- Cadence
- Simulation Tools

Outcome:

Developed a neuromorphic system model demonstrating efficient learning capability and reduced power consumption, suitable for next-generation AI hardware systems.

Skills and Certifications:

Technical Skills (VLSI & Core Electronics:

- Strong understanding of CMOS technology and digital VLSI design fundamentals
- Knowledge of CMOS design, logic gates, and transistor-level circuit design
- Familiarity with low-power design techniques and circuit optimization methods
- Understanding of analog and mixed-signal basics relevant to VLSI systems
- Knowledge of VLSI design flow including simulation and verification

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Neuromorphic & Emerging Technology Skills:

- Understanding of Leaky Integrate-and-Fire (LIF) neuron modeling
- Knowledge of memristor based circuit design
- Familiarity with Spike-Timing Dependent Plasticity (STDP) learning mechanisms
- Exposure to neuromorphic computing and brain-inspired hardware design

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Tools & Practical Skills:

- Circuit design and simulation (e.g., LTspice / Cadence – mention what you used)
- Basic experience with HDL (Verilog/VHDL – if applicable)

- Schematic design and analysis
- Problem-solving in electronic circuit design

Research & Academic Skills:

- Technical documentation and report writing
- Research paper preparation and academic formatting
- Analytical thinking and system-level understanding
- Ability to translate theoretical concepts into practical implementations

Certifications:

• I have successfully completed an NPTEL course in CMOS VLSI Digital Design, gaining a strong foundation in CMOS technology, logic design, and circuit optimization techniques.

- I presented and published my research work at the IC-AISmart, which is indexed in Scopus, and received a certificate for my contribution. This experience improved my research, technical writing, and presentation skills, and strengthened my interest in pursuing higher studies in VLSI and advanced technologies.

Internship:

Title: PCB Design and Fabrication

VTU Internship – Employability Skills Program

- Completed training focused on professional and technical skill development
- Learned communication, problem-solving, and industry-oriented skills
- Gained hands-on experience in PCB Design using KiCad

Skills:

- Technical Skills:
- VLSI Basics
- CMOS Fundamentals
- Analog & Digital Circuit Design
- Embedded Systems
- PCB Fabrication
- Neuromorphic Systems Basics

Tools:

- Cadence
- KiCad
- Simulation Tools

Soft Skills

- Problem-solving
- Teamwork

Conclusion:

- During my internship in PCB design and fabrication, I gained practical experience in schematic design, layout development, and understanding the complete PCB manufacturing process.
- I developed skills in component placement, routing, and design considerations for reliable and efficient circuits.
- This experience strengthened my hands-on knowledge in hardware design and reinforced my interest in VLSI and electronic system development.